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Prolonged Initial Empirical Antibiotic Treatment is Associated with Adverse Outcomes in Premature Infants

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Objective To investigate the outcomes after prolonged empirical antibiotic administration to premature infants in the first week of life, and concluding subsequent late onset sepsis (LOS), necrotizing enterocolitis (NEC), and death.

Study design Study infants were ≤32 weeks gestational age and ≤1500 g birth weight who survived free of sepsis and NEC for 7 days. Multivariable logistic regression was conducted to determine independent relationships between prolonged initial empirical antibiotic therapy (≥5 days) and study outcomes that control for birth weight, gestational age, race, prolonged premature rupture of membranes, days on high-frequency ventilation in 7 days, and the amount of breast milk received in the first 14 days of life.

Results Of the 365 premature infants who survived 7 days free of sepsis or NEC, 36% received prolonged initial empirical antibiotics, which was independently associated with subsequent outcomes: LOS (OR, 2.45 [95% CI, 1.28-4.67]) and the combination of LOS, NEC, or death (OR, 2.66 [95% CI, 1.12-6.3]).

Conclusions Prolonged administration of empirical antibiotics to premature infants with sterile cultures in the first week of life is associated with subsequent severe outcomes. Judicious restriction of antibiotic use should be investigated as a strategy to reduce severe outcomes for premature infants. *(J Pediatr 2011;159:720-5)*.

See editorial, p 707

Antibiotics are the most commonly prescribed medications for newborns in intensive care nurseries.[1] In the United States, the majority of the very premature infants receive empirical antibiotic treatment during the first days of life even though the incidence of early-onset sepsis (EOS) is low.[2,3] Concerns about occult intrauterine infection that precipitates premature labor, premature rupture of membranes, and chorioamnionitis often prompt initiation of empirical antibiotic treatment.[4] Although antibiotic treatment of premature infants may be prudent given these considerations, the duration of treatment is often arbitrary, based not on positive culture results but on the clinician's perceived risk of infection.[5] Intensive broad-spectrum antibiotic use can have serious, unintended consequences in premature infants, including rapidly increasing drug resistance in sepsis cases[6] and increased risk of invasive fungal infection.[7]

Antibiotic therapy may have significant adverse consequences in early postnatal weeks,[8,9] which coincides with the time of initial gastrointestinal colonization.[10,11] Although it is not known to what extent antibiotic exposure disrupts colonization of the developing infant gastrointestinal tract, preterm infants have intestinal microflora distinct from that of healthy, full-term infants.[12,13] Moreover, several observations suggest the importance of gastrointestinal colonization to the health of premature infants. Results of randomized controlled trials that provided probiotic organisms to preterm infants have reported decreased adverse outcomes, including late-onset sepsis (LOS), necrotizing enterocolitis (NEC), and death.[14,15] Human milk contains prebiotic oligosaccharides that stimulate beneficial colonization of the gastrointestinal tract,[16] and provision of human milk reduces the incidence of LOS and NEC among premature infants.[17-19]

We tested the hypothesis that prolonged initial empirical antibiotic use in preterm infants is an independent risk factor for development of the combination of outcomes that appear related to aberrant colonization: LOS, NEC, or death. Human milk feeding and other clinical factors were examined and controlled for as potential confounders.

ELBW	Extremely low birth weight
EOS	Early-onset sepsis
LOS	Late-onset sepsis
NEC	Necrotizing enterocolitis
NICHD NRN	*Eunice Kennedy Shriver* National Institute of Child Health and Human Development Neonatal Research Network

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Methods

We conducted a retrospective cohort study of 365 very low birth weight infants (≤32 weeks gestational age and ≤1500 g birth weight) who survived the first week of life without sepsis and NEC. All the infants were part of a cohort originally identified at 3 neonatal intensive care units in Cincinnati, from April 2000 through December 2004, to examine epidermal growth factor in relation to NEC.[20] Infants in the original cohort excluded from this study included 8 infants who died, 11 infants with culture-proven sepsis, and 2 infants with NEC in the first week of life. Nineteen other infants in the original cohort who did not receive empirical antibiotic treatment on day of life one but who underwent an evaluation for sepsis and received antibiotic treatment within the first week of life were excluded. Study infants were participants in the *Eunice Kennedy Shriver* National Institute of Child Health and Human Development Neonatal Research Network (NICHD NRN) registry. As part of the NICHD NRN system, infants were enrolled immediately after delivery and were followed up until discharge, transfer, 120 days postpartum, or death. In addition to the standard NRN data, supplemental maternal and infant data were abstracted from medical records. Institutional review boards at Cincinnati Children's Hospital Medical Center, Good Samaritan Hospital, and University Hospital approved this study.

We defined initial empirical antibiotic treatment as the antibiotic treatment initiated within the first postnatal day. The duration of initial empirical antibiotic therapy was defined as the total number of continuous days of administration of antibiotics with sterile culture results. The infants were categorized into 3 groups: 0 days of initial antibiotic therapy, 1-4 days of initial antibiotic therapy (limited antibiotics), and ≥5 days of initial antibiotic therapy (prolonged antibiotics). The cumulative days of antibiotic treatment also were collected for the hospital course. EOS was defined as in the NICHD NRN registry on the basis of positive blood culture results obtained within the first 3 postnatal days and treated for ≥5 days.[2,3] Infants diagnosed with EOS were excluded from the analysis of empirical antibiotic treatment. LOS was defined as a positive blood, cerebrospinal fluid, urine, or sterile site culture after 3 postnatal days. Coagulase-negative staphylococci and polymicrobial cultures were included as LOS if the clinical team indicated it was a true infection in the medical record and that they treated the infection accordingly. NEC was defined by using modified Bell stage II or III criteria.[21] "Full enteral feeds" was defined as successful intake of at least 120 mL/kg/d.

Analysis

Given the fixed sample size of 365 very low birth weight infants, the combined outcomes of LOS, NEC or death (91 cases), and LOS alone (76 cases) were used as the primary outcome variables to assure 80% power to detect at least a 2-fold association between prolonged and shorter duration antibiotic use groups. The number of deaths (20 cases) and NEC alone (17 cases) was too few to provide sufficient power to analyze independently. Factors associated with the median duration of antimicrobial treatment and the frequency of prolonged antibiotic treatment in the first week of life were examined for all infants. Maternal and neonatal baseline characteristics collected in the first 7 postnatal days were compared across these 3 groups. Differences among the 3 antibiotic groups were tested by using the χ^2 test of proportions for categorical variables and analysis of variance for continuous variables. When further comparisons were made between the 2 groups, a multiple comparison correction was made. The Wilcoxon rank sum test was used when comparisons of median values were tested. Median values for continuous variables were reported when the data did not follow a normal distribution.

Multivariable logistic regression models were used to evaluate independent associations between prolonged initial empirical antibiotic treatment and the primary outcomes of LOS, NEC, or death, and LOS alone, and secondary outcomes of NEC alone and death alone. LOS, NEC, or death outcomes that occurred on or after day of life 7 were used in the analyses. In addition to prolonged initial empirical antibiotic treatment, a comprehensive set of clinical predictors of mortality identified from a previous national study were examined in the models.[22] Final regression models were parsimonious and included only covariates significant at the .10 level. The Hosmer-Lemeshow test statistic and other model diagnostics were used to select the final models based on the best fit.

Results

Clinical and demographic characteristics of study infants and their mothers segregated by initial empirical antibiotic treatment group are presented in **Table I**. Ampicillin and gentamicin were the universally prescribed antibiotic combination for initial empirical treatment. Other antibiotics also prescribed during the first 7 days of life included clindamycin (1.4%), amphotericin B (1%), nafcillin (0.8%), and cefotaxime (0.8%). One infant received erythromycin secondary to isolation of Ureaplasma from amniotic fluid. Infants in the prolonged antibiotic group were significantly more likely to have lower birth weight, younger gestational age, an Apgar score at 5 minutes of age less than 6, endotracheal intubation, and surfactant treatment for clinical features of respiratory distress syndrome, more conventional ventilation days, high-frequency ventilation days, and highest oxygen (fraction of inspired oxygen) concentration at 7 days of life compared with infants in the limited antibiotic groups. Infants in the prolonged antibiotic group were more likely to have a mother diagnosed with chorioamnionitis and a mother who received antepartum antibiotics compared with infants in the limited antibiotic groups. However, infants in the prolonged antibiotic group were less likely to have a mother with hypertension or eclampsia, to have been the product of a multiple gestation, or to have been delivered by cesarean section.

Table I. Characteristics of study infants by initial empirical antibiotic use

	Initial empirical antibiotic therapy, n = infants (% of total study infants)			
Variable, description	0 Day, n = 60 (16.4%)	1-4 Days, n = 175 (48%)	≥5 Days, n = 130 (35.6%)	*P* value
Infant baseline characteristics				
Birth weight, mean (SD) (g)+	1203 ± 227	1069 ± 276	935 ± 253*	<.001
Gestational age, mean (SD) (wk)+	30.4 ± 1.9	28.2 ± 2.2	26.5 ± 2.1*	<.001
Non-Hispanic black, no. (%)+	11 (18.3)	52 (29.7)	43 (33)	.11
Male, no. (%)^	24 (40)	76 (43.4)	66 (50.8)	.29
Maternal and prenatal information				
Prenatal steroids given (any), no. (%)+	55 (91.7)	154 (88)	113 (86.9)	.64
Hypertension/eclampsia, no. (%)+	33 (55)	43 (34.6)	13 (10)*	<.001
Prenatal care (≥1 prenatal visit), no. (%)+	60 (100)	168 (96)	123 (94.6)	.19
Married, no. (%)+	40 (66.7)	106 (60.6)	67 (51.5)	.10
Antepartum antibiotic therapy, no. (%)	32 (53.3)	108 (61.7)	102 (78.5)*	<.001
Multiple birth, no. (%)^	20 (33)	67 (38.3)	32 (24.6)*	.04
Prepartum hemorrhage, no. (%)^	6 (10)	31 (17.7)	28 (21.5)	.15
Labor and delivery information				
Cesarean section delivery, no. (%)	46 (76.7)	123 (70.3)	71 (54.6)*	.003
Chorioamnionitis, no. (%)	1 (1.7)	10 (5.8)	58 (47.9)*	<.001
Rupture of membranes > 24 h, no. (%)	6 (10)	36 (20.6)	34 (26.2)	.04
Delivery room resuscitation, no. (%)	40 (66.7)	158 (90.3)	123 (94.6)	<.001
Type of delivery room resuscitation				
Positive pressure ventilation, no. (%)	37 (92.5)	124 (78.5)	45 (36.6)*	<.001
Endotracheal intubation, no. (%)	4 (10)	35 (22.2)	79 (64.2)*	<.001
Epinephrine, no. (%)	0	7 (4.4)	7 (5.7)	.31
Chest compressions, no. (%)	0	8 (5.1)	11 (8.9)	.09
5-min Apgar score of 3-6, no. (%)+	0	33 (19.9)	35 (26.9)*	<.001
5-min Apgar score of > 6, no. (%)+	60 (100)	137 (78.3)	79 (60.8)	
Received surfactant, no. (%)	5 (8.3)	84 (48)	99 (76.2)*	<.001
Clinical features of respiratory distress syndrome, no. (%)+	35 (58.3)	162 (92.6)	126 (96.9)	<.001
Infant information to day 7				
Median no. days (range) with conventional ventilation at day 7^	0 (0-5)	0 (0-7)	2 (0-7)*	<.001
Median no. days (range) with high-frequency ventilation at day 7^	0 (0-5)	0 (0-4)	0* (0-7)	<.001
Highest oxygen (FiO_2) concentration at day 7, median (range)^	0.21 (0.21-0.70)	0.21 (0.21-1.0)	0.35 (0.21-1.0)*	<.001

FiO_2, fraction of inspired oxygen.
Clinical factors previously associated with higher survival rates (+) or associated with a higher risk of death (^) from birth to day 7 of life that had complete data for analysis in this study.[22]
*Comparisons between infants who received antibiotics 1-4 vs infants who received antibiotics ≥5 days statistically significant.

Initiation and Advancement of Enteral Nutrition

Among neonatologists in Cincinnati, the consensus practice was to initiate enteral feeding within the first few days of life and to advance feeding volume in a range between 10 mL/kg/d and 20 mL/kg/d by using the mother's own breast milk when available. Given the preference for early feeding, delayed initiation or advancement of feeding reflected clinical perception of increased severity of illness. The time to initiation of enteral feeding (median [range], 3 days [0-18 days] vs 2 days [0-36 days]; $P < .001$) and to attainment of full enteral feeding (16 days [4-80 days] vs 11 days [1-65 days]; $P < .001$) was delayed for infants who received prolonged initial empirical antibiotic treatment compared with those who received limited initial antibiotics. About a fourth of the infants received no human milk in the first 14 days of life and this proportion did not differ among groups according to early empirical antibiotic treatment duration. The total human milk intake over the first 14 days of life was significantly lower ($P < .001$) among infants who received prolonged initial empirical antibiotics (median [range], 131 mL/kg [0-1440 mL/kg]) compared with those who received limited initial empirical antibiotics (345 mL/kg [0-1600 mL/kg]). Similarly, the number of infants who received ≥500 mL/kg of human milk in the first 14 days of life was significantly lower ($P < .001$) among infants treated for with prolonged initial empirical antibiotics (22%) compared with infants who received limited initial empirical antibiotics (40%).

LOS, NEC, or Death Outcomes

A total of 76 study infants (21%) were diagnosed as having LOS, 17 (4.6%) were diagnosed with NEC (Bells stage ≥ 2), and 20 infants (5.5%) died. LOS, NEC, or death occurred more often in the prolonged antibiotic group compared with the limited antibiotic group (Table II). Also, LOS alone and death but not NEC alone occurred more often in the prolonged antibiotic group compared with the limited antibiotic group infants. A total of 122 pathogens were isolated from 100 positive cultures (82 gram-positive bacteria, 30 gram-negative bacteria, and 10 *Candida* isolates) from these 76 infants. Pathogens included *Candida*, *Citrobacter*, *Escherichia coli*, *Enterobacter*, *Enterococcus*, group *β Streptococcus and other Streptococcal* species, *Klebsiella*, *Pseudomonas*, *Serratia*, *Staphylococcus aureus*, *and* coagulase-negative *Staphylococcus*. Coagulase-negative *Staphylococcus* was isolated in a half of LOS cases. Case fatality was higher in gram-negative (3 deaths in 30 cases [10%]) than gram-positive sepsis (5 deaths in 82 cases, 6.1%). Multivariable analyses (Table III) found that prolonged early antibiotic therapy was independently associated with the composite outcome of LOS, NEC, or

Table II. Initial empirical antibiotics, total antibiotic exposure, and select adverse outcomes

	Initial empirical antibiotic therapy, n = infants (% of total study infants)			
Description	0 Days, n = 60 (16.4%)	1-4 Days, n = 175 (48%)	≥5 Days, n = 130 (35.6%)	*P* value
Total days treated with antibiotics during hospital course, median (range) [interquartile range]	0 (0-21) [0-0]	3 (1-55) [3-8]	14 (5-84) [8-27]†	<.0001
Composite, no. (%)*	7 (11.7)	31 (17.7)	53 (40.8)†	<.0001
Late onset sepsis, no. (%)	7 (11.7)	23 (13.1)	46 (35.4)†	<.0001
NEC, no. (%)	0	8 (4.6)	9 (6.9)	.11
Death, no. (%)	0	8 (4.6)	12 (9.2)	.03

*Any outcome of death, sepsis, or NEC after 7 days of life.

death, and with LOS after 7 days of life controlling for birth weight, gestational age, race, prolonged premature rupture of membranes, number of days on high-frequency ventilation in first week of life and the amount of breast milk received in the first 14 days of life. For each day of initial empirical antibiotic therapy, the odds of LOS, NEC, or death, and of developing LOS increased significantly. For infants who had received any initial empirical antibiotic exposure (305 infants), the adjusted attributable risk of LOS, NEC, or death was 32 per 100 infants, and the number needed to harm was 3.

Discussion

We found that prolonged initial empirical antibiotic therapy is associated with a 2-fold higher incidence of LOS, NEC, or death, and with a 3-fold higher incidence of LOS alone. Importantly, these associations persisted after adjusting for proxy severity of illness indicators previously identified as predictors for mortality.[22] Although administration of empirical antibiotics to preterm infants is common and is regarded to be safe, our study and others suggest that the duration of therapy is often arbitrary and the risk-benefit ratio of prolonged empirical antibiotic use may in fact be unfavorable. Cordero and Ayres[5] reported that, in their cohort of 742 extremely low birth weight (ELBW) (≤1000 g birth weight) infants with initial sterile blood cultures 60% received empirical antibiotics for >3 days. They concluded that the decision to extend the duration of empirical antibiotic therapy appeared to be an institutional decision not one based on severity of illness. Cotten et al[8] reported that 53% of the 4039 ELBW infants enrolled in the NICHD NRN between 1998 and 2001 received initial empirical antibiotic therapy for ≥5 days' duration despite sterile blood cultures. In their study, prolonged initial empirical antibiotic treatment of ELBW infants was associated with significantly higher rates of death and NEC or death, and a trend toward a higher rate of NEC alone even after adjusting for study center, gestational age, and other perinatal factors. A supplemental analysis of their data showed that LOS or death was significantly associated with prolonged initial empirical antibiotic therapy when organisms other than coagulase-negative *Staphylococcus* caused LOS. Although Cotten et al[8] did not address confounding by infant feedings, we found that, even though human milk feeding was independently protective, it did not confound the relationship between antibiotic therapy and adverse outcomes.

Limitations of this study are the inability to control for all factors that indicate severity of illness. Although we sought to enhance the characterization of severity of illness by using established respiratory indices through day of life 7, this eliminated potential study subjects with early LOS, NEC, or death. As a result our study, the cohort size was reduced, which limited the power to detect potential differences in

Table III. Multivariable logistic regression for 305 infants who had received any empirical antibiotic therapy

	OR	95% CI	*P* value
Composite outcome: NEC, LOS, death after DOL 7*			
Initial empirical ABX duration per day	1.24	1.07-1.44	.005
Initial empirical ABX duration ≥ 5 days	2.66	1.12-6.30	.016
Outcome: LOS after DOL 7*			
Initial empirical ABX duration per day	1.27	1.09-1.49	.003
Initial empirical ABX duration ≥ 5 days	2.45	1.28-4.67	.007
Outcome: NEC after DOL 7†			
Initial empirical ABX duration per day	1.08	0.83-1.40	.57
Initial empirical ABX duration ≥ 5 days	1.28	0.42-3.93	.66
Outcome: death after DOL 7‡			
Initial empirical ABX duration per day	1.04	0.82-1.33	.74
Initial empirical ABX duration ≥ 5 days	1.12	0.40-3.10	.83

DOL, day of life; *ABX*, antibiotic therapy.
*Controlling for birth weight, gestational age, race, prolonged premature rupture of membranes, number of days on high frequency ventilation in first week of life, amount of breast milk received in first 14 days of life.
†Controlling for gestational age, race, number of days on mechanical ventilation in first week of life, amount of breast milk received in first 14 days of life.
‡Controlling for birth weight, race, 5-minute Apgar < 5, amount of breast milk received in first 14 days of life.

death alone or NEC alone. In addition, elimination of infants who developed early sepsis limited the utility of early leukocyte indices and acute phase reactants in identifying culture-negative infants that might benefit from a prolonged duration of antibiotics.

Prolonged initial treatment with antibiotics may be indicated for preterm newborns when the likelihood of sepsis is high; however, the broad use of prolonged antibiotics might impair important transitional events required for intestinal homeostasis. Antibiotic therapy is known to alter colonization of the gastrointestinal tract and predispose to the emergence of pathogens and resistant organisms. Animal studies of intestinal development underscore abnormal interaction between the intestinal epithelium and the luminal microbial milieu when colonization is interrupted by housing animals in germ-free conditions or treating them with broad-spectrum antibiotics.[23,24] Activation of toll-like receptors by commensal bacteria appears to be critical for protection against gut injury and associated mortality.[25] The lack of bacterial species diversity and abundance of Proteobacteria species associated with widespread use of antibiotics may predispose to inflammatory stimulation that may help explain the susceptibility of premature newborns to LOS and NEC.[13,26]

Results of our study suggest that there may be inherent risks associated with initial empirical antibiotic treatment and that each day of antibiotic therapy increases the odds of severe outcomes. Although it may be prudent to discontinue antibiotics for many premature infants with suspected EOS when cultures are negative, we do not propose limiting antibiotic treatment duration in all premature infants because blood-sampling limitations and maternal antepartum antibiotic coverage may preclude optimal culture sensitivity.[27] We do advocate rigorous review of ongoing empirical antibiotic treatment and prompt discontinuation of therapy if blood cultures are negative, and clinical and laboratory measurements indicate a low risk of sepsis. Automated blood culture systems now provide positive culture results within 48 hours after initiation of culture.[28] Also, adjuvant diagnostic tests such as leukocyte indices, serum acute phase reactants, leukocyte cell surface markers, and certain proinflammatory cytokines may be used to estimate sepsis risk or to determine adequacy of treatment, thus allowing discontinuation of antibiotic therapy.[29,30] Microbial molecular genetic methods under development will likely transform clinical practice early, sensitive detection of microbial pathogens and thus, prompt discontinuation of antibiotic therapy when test results are negative.[29,31] Finally, molecular genetic techniques now being applied to characterize the developing intestinal microbiota will enhance our understanding of the effects of antibiotic exposure on intestinal colonization and predisposition of premature infants to LOS, NEC, and death.[13] Such knowledge may inform rational therapeutic use of antibiotics, prebiotics, and probiotics.

The data from this and other retrospective studies suggest that prolonged initial empirical antibiotic treatment may predispose to adverse outcomes in premature infants. Adopting a standardized approach to empirical antibiotic prescription by using clinical indicators, adjuvant diagnostic tests, and molecular techniques may enable clinicians to achieve more judicious use of antimicrobials by using predefined clinical criteria. Prospective trials are needed to determine whether judicious restriction of early antibiotic treatment might reduce the risk of LOS, NEC, and death for premature infants. ■

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